

Asia Pacific Scenarios Development Global Environmental Outlook

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Outline of the Presentation

- 1) Background Information on GEO Scenario Development**
- 2) Narrative GEO Scenarios (Asia Pacific Specific Storylines)**
- 3) Quantitative Illustration of GEO Scenarios**
- 4) Illustration of SRES Scenarios**

1) Background Information on GEO Scenario Development

- GEO - 1 *(Chapter: Looking to the Future)*
- GEO - 2000 *(Chapter: Outlook)*
- GEO - 3 *(Chapter: Outlook)*

❖ ***Key Features***

- **Analysis at regional and sub-regional levels**
- **Major driving forces**
 - *Demographics* - *Economics*
 - *Social* - *Culture*
 - *Technology* - *Environment*
 - *Governance*
- **Time horizon: 2002-2032**

Comparison of Scenario Framework

- Global Scenario Group (GSG)
- Special Report on Emission Scenarios (SRES)
- World Business Council on Sustainable Development (WBCSD)
- **GEO-3 Scenarios**
 - Different time horizon
 - Focuses on environmental issues in addition to climate change
 - Considers poverty alleviation and equity issues
 - Storylines include themes such as social, cultural and governance

Comparison of Scenarios (contd.)

Framework

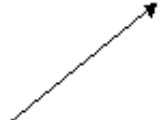
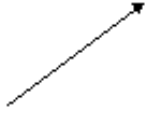
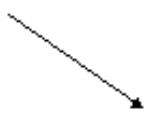
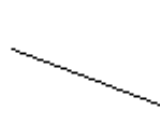
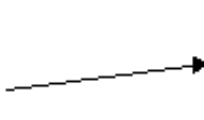
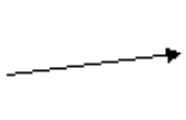
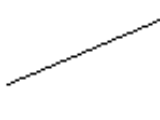
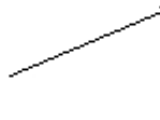
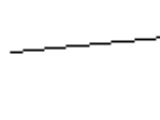
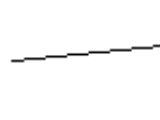
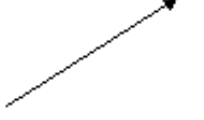
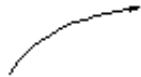
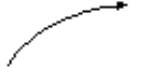
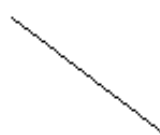
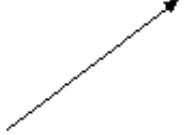
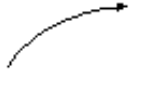
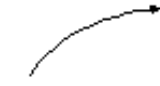
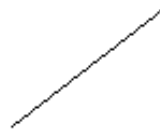
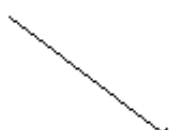
Scenarios

GSG	SRES	WBCSD	GEO-3
Conventional Worlds <i>Market Forces</i> <i>Policy Reforms</i>	A1 B1	FROG! Geopolity	Market Forces Policy Reforms
Barbarisation <i>Breakdown</i> <i>Fortress world</i>	A2		Fortress world
Great Transitions <i>Eco-communalism</i> <i>New sustainability paradigm</i>	B2	Jazz	Great Transitions

2) Narrative GEO Scenarios (Asia Pacific Specific Storylines)

- **Four Scenarios:**
 - Market Forces Scenario
 - Policy Reform Scenario
 - Fortress World Scenario
 - Great Transitions Scenario
- **For first two scenarios, quantitative illustrations are provided for 1995 and two future years: 2015 and 2032**
(in GEO Scenario Framework: UNEP, 2004)

Scenarios Structure with Illustrative Patterns of Change

Scenario	 Population	 Economy	 Environment	 Equity	 Technology	 Conflict
Market forces						
Policy Reform						
Fortress World						
Great Transitions						

Source: GEO Scenario Framework, UNEP 2004

GEO-4 Meeting, November 18-19, 2004 Bangkok, Thailand

Market Forces Scenario

(Selected Asia Pacific Storylines)

- Rapid increase in population in East and South Asia
- South Asia experiences accelerated economic growth rates
- Urban environment continues to degrade in most large cities
- Development and increased use of cleaner technologies (solar, wind, micro-hydro, bio-gas, etc.)
- Scarcity of fresh water increases
- Increase in transboundary competition for key non-renewable natural resources

Policy Reform Scenario

(Selected Asia Pacific Storylines)

- Integrated economic and environmental policies, environmental pricing and phasing out of subsidies to environmentally harming activities are undertaken.
- Implementation of policies related to education, distribution of basic needs such as water and power, and health
- Policies to promote environment friendly technologies in the field of energy and IT
- Natural forests preserved, and reforestation initiatives taken
- Development Assistance supports implementation of environment-related policies in less developed countries

Fortress World Scenario

(Selected Asia Pacific Storylines)

- Collapse of major “powerhouses” of economic and technological growth in the Northwest Pacific and Southeast Asia
- Increased protectionist policies in countries
- Greater pressure on domestic natural resources due to reduction in imports of natural resources, especially oil and gas
- Water resource stress, land degradation and solid waste accumulation are the critical environmental problems.
- Corruption and breakdown of regulatory mechanisms, complete breakdown of governance, weak rule of law and prevalence of terrorism and militancy are prevalent.
- Highly populated countries like India adopt one-child policy

Great Transition Scenario

(Selected Asia Pacific Storyline)

- Asia eventually emerges as a powerhouse within the world
- Energy- related technological progress sees the reduction of energy consumption, as well as SO_x and CO₂ emission
- Equitable access to basic services (potable water and sanitation), IT services and literacy rates, extending to over 80 per cent
- Income sharing policies (fiscal, taxation and market based) see more equitable distribution of wealth and spread of benefits from the powerhouse nations to the wider region.
- Population growth, structure and density will stabilize
- Gradual improvement in the quality of the environment due to enhanced role of civil society and business in governance

3) Quantitative Illustration of GEO Scenarios

Key indicators with numerical summaries in GEO scenarios

Demography

Population

Urbanization

Economy and Society

GDP (PPP)

GDP in Industry, Services, Agriculture

Income (GDP per capita)

Hunger Incidence (% of population)

National Equity (Low 20%/High 20%)

Energy

Primary energy

Primary fuel shares Energy intensity

Final fuel demand

Sector share of final fuel demand

Food and Agriculture

Average daily consumption

Meat & milk requirements

Crop requirements

Fish requirements

Irrigated area

Cereal yield and cropping intensity

Environmental Pressure

Land use

Water use, Water stress

CO2 emissions

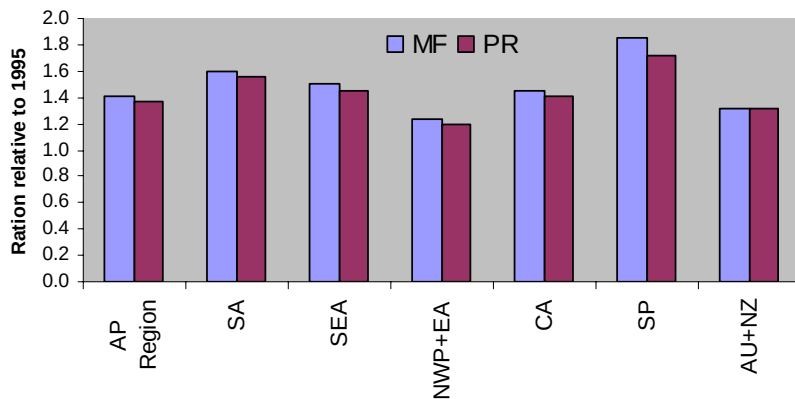
Sulfur emissions

Municipal solid waste generation

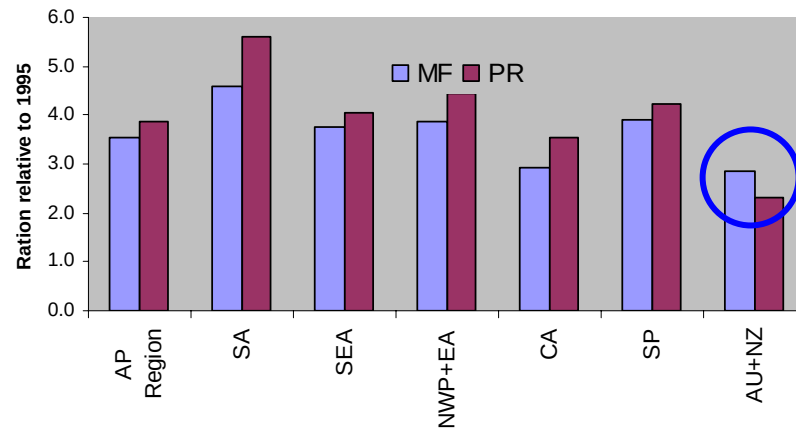
Toxic waste

Demographic and economic indicators under MF and PR

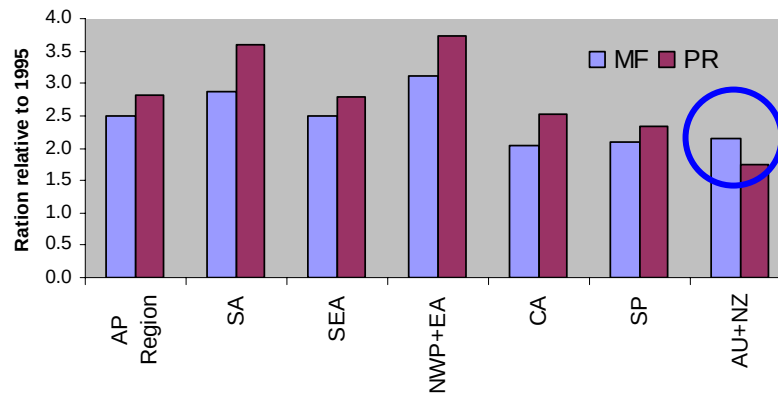
Population in Asia Pacific Region under MF and PR Scenarios, 2032



GDP in Asia Pacific Region under MF and PR Scenarios, 2032

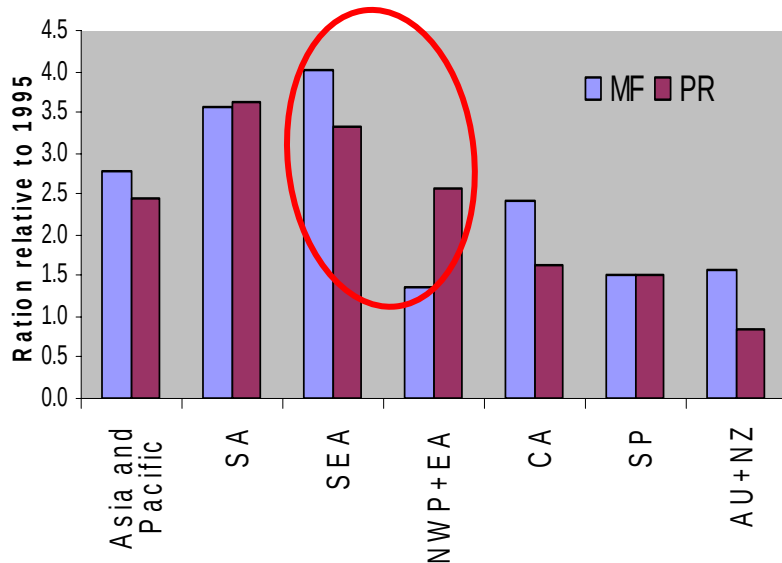


GDP per Capita in Asia Pacific Region under MF and PR Scenarios, 2032

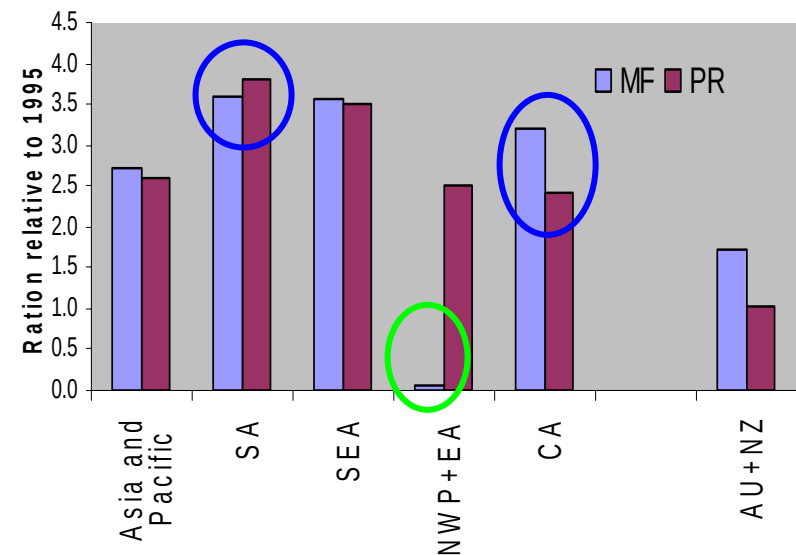


Energy indicators under MF and PR

Primary Energy Requirement in Asia Pacific Region under MF and PR Scenarios, 2032

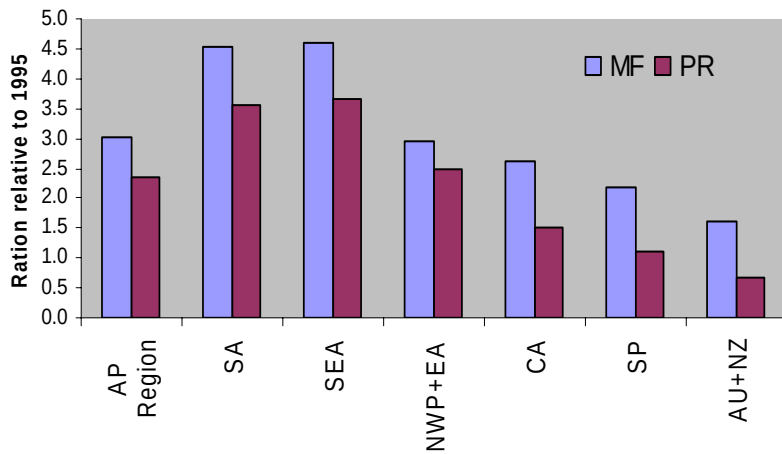


Final Energy Demand in Asia Pacific Region under MF and PR Scenarios, 2032

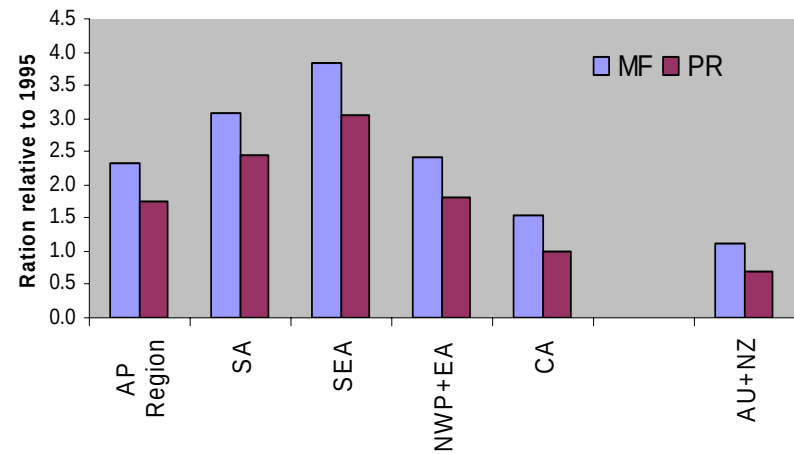


Environmental indicators under MF and PR

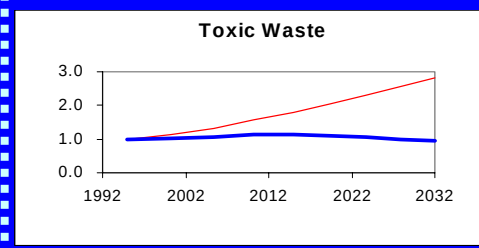
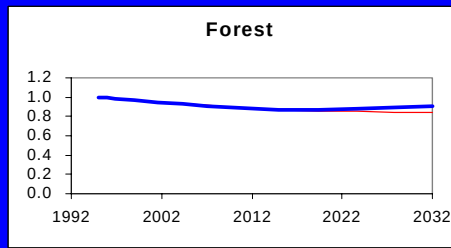
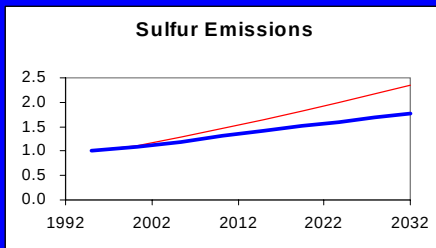
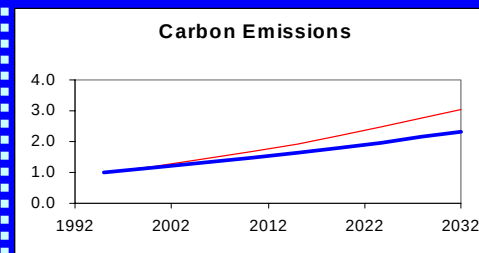
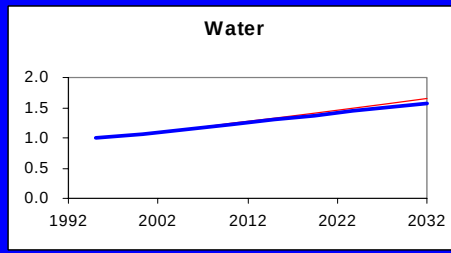
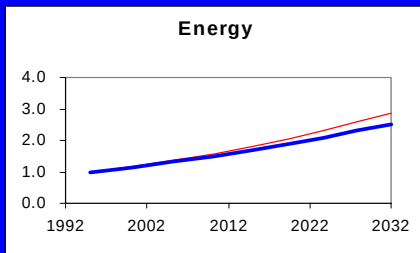
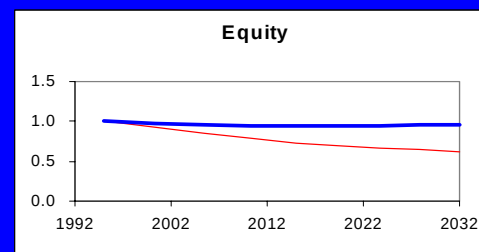
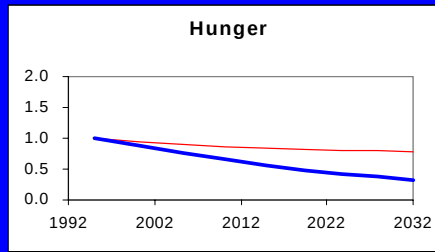
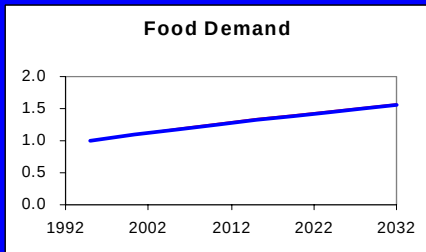
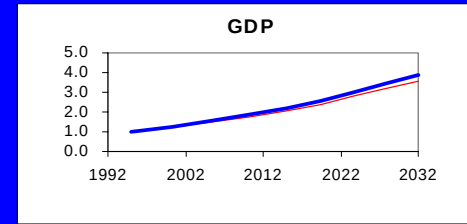
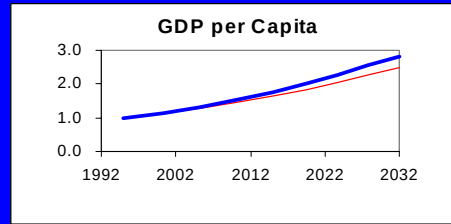
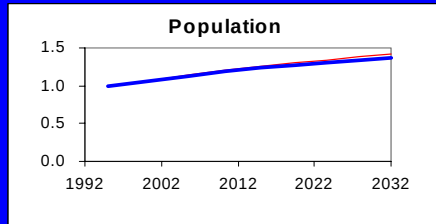
Carbon Emission in Asia Pacific Region under MF and PR Scenarios, 2032



Sulfur Emission in Asia Pacific Region under MF and PR Scenarios, 2032



Graphical representation of key indicators



MF

PR

Values index
to 1 in 1995

4) ILLUSTRATION OF SRES SCENARIO *(With the application of AIM)*

APPROACH TO SCENARIO DEVELOPMENT

General

- **TASK 1:** Collect Information about the issues and prepare narrative description
- **TASK 2:** Translate narrative description to systematic and quantitative data structure

Specific to energy and environment

- **TASK 3:** Prepare storylines and socio-economic scenarios
- **TASK 4:** Simulate future environmental pressures and analyze effect of strategies
 - Affects model parameters
 - Identify extent of effect

SRES Socio-Economic Scenarios

Market integration

		Integrated	Fragmented
<i>Governance</i>	Centralization	A1 Fast Growth	A2 Pre-reform (Mixed Economy Model)
	Decentralization	B1 Sustainable Development	B2 Self Reliance Model

Source: Shukla et. al (2004)

Scenarios, Key drivers and Impact on Model Parameter

A1

➤ High globalization and growth

➤ **Key Drivers:** competition and private participation, access to global finance and technology, technology R&D, transfer and capacity building, reduced risk perception

➤ Implication on critical parameters

Fuel price (↑), technology cost (↓), efficiency (↑), T&D losses (↓), investment capacity (↑), technology choices (↑), discount rate (↓) etc

A2

➤ Pre-reform (Mixed-economy model)

➤ **Key Drivers:** GDP growth, energy efficiency, Non-fossil fuels vs fossil fuels, oil consumption, technological change, movement on the fuel ladder

➤ Implication on critical parameters

Sectoral demand (↑ ↓), investment limit (↑ ↓), Fuel supply (↑ ↓), fossil-fuel price (↑ ↓) etc.

B1

➤ Sustainable Development

➤ **Key Drivers:** Strong environmental awareness and values, consumption changes, dematerialization, cooperation, shift away from fossil fuels, capacity building, rural energy and electricity development

➤ Implication on critical parameters

Environmental constraints (↑), energy content of goods/services (↓), electricity consumption due to efficiency improvements (↓), T&D losses (↓), penetration and CRT (↑), fossil-fuel prices (↑), organic fertilizer use (↑) etc.

B2

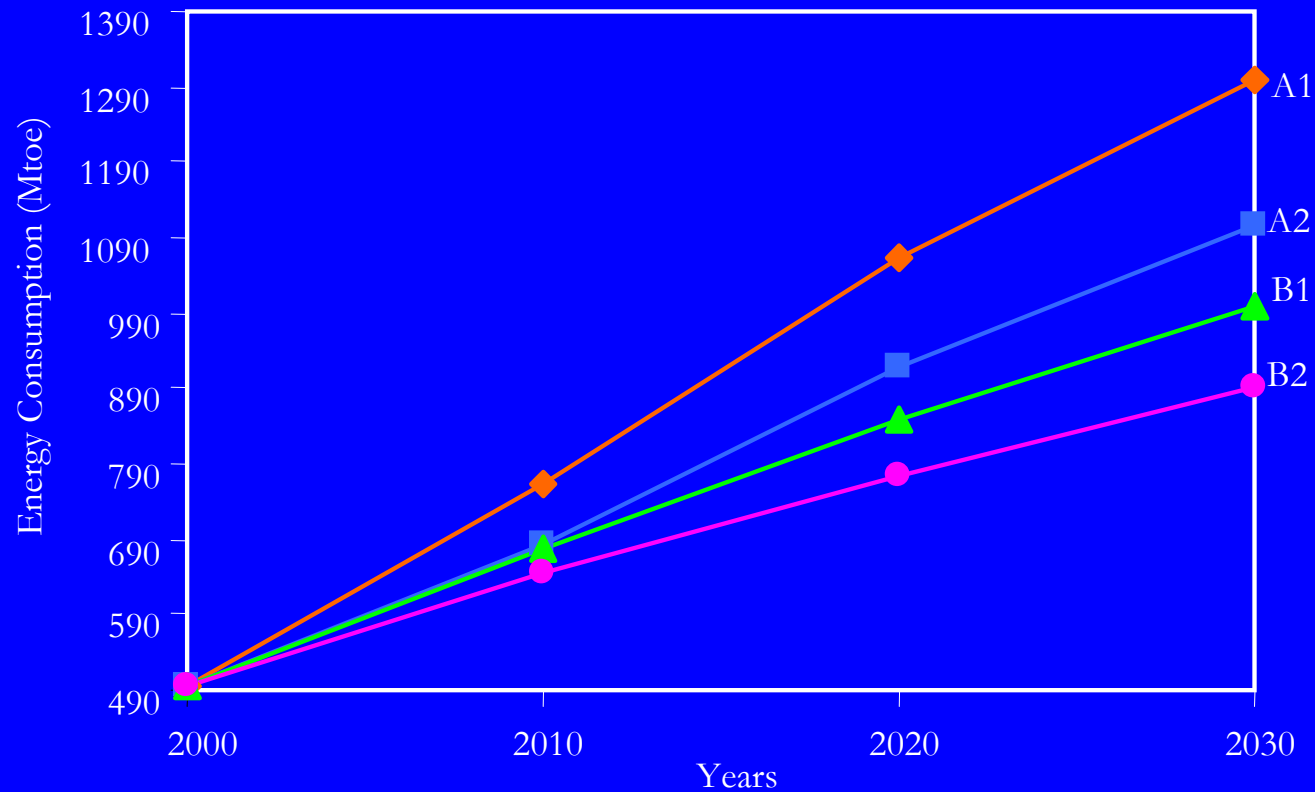
➤ Regional development

➤ **Key Drivers:** GDP growth, population, local emissions, productivity, importance of agriculture, resource intensity, self-reliance

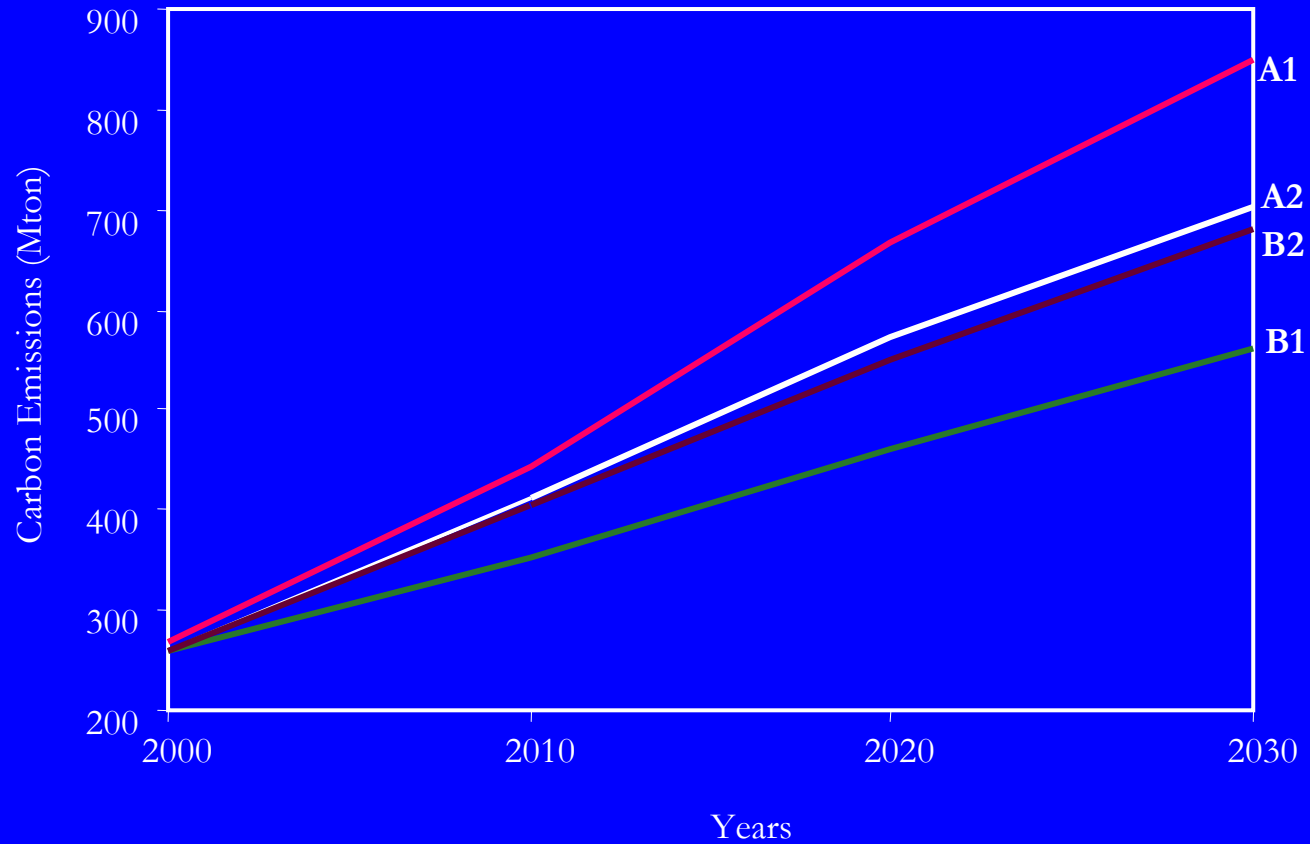
➤ Implication on critical parameters

Population growth (↑), sectoral demand (↑ ↓), investments (↓), penetration of CRT (↑), agricultural share (↑) etc.

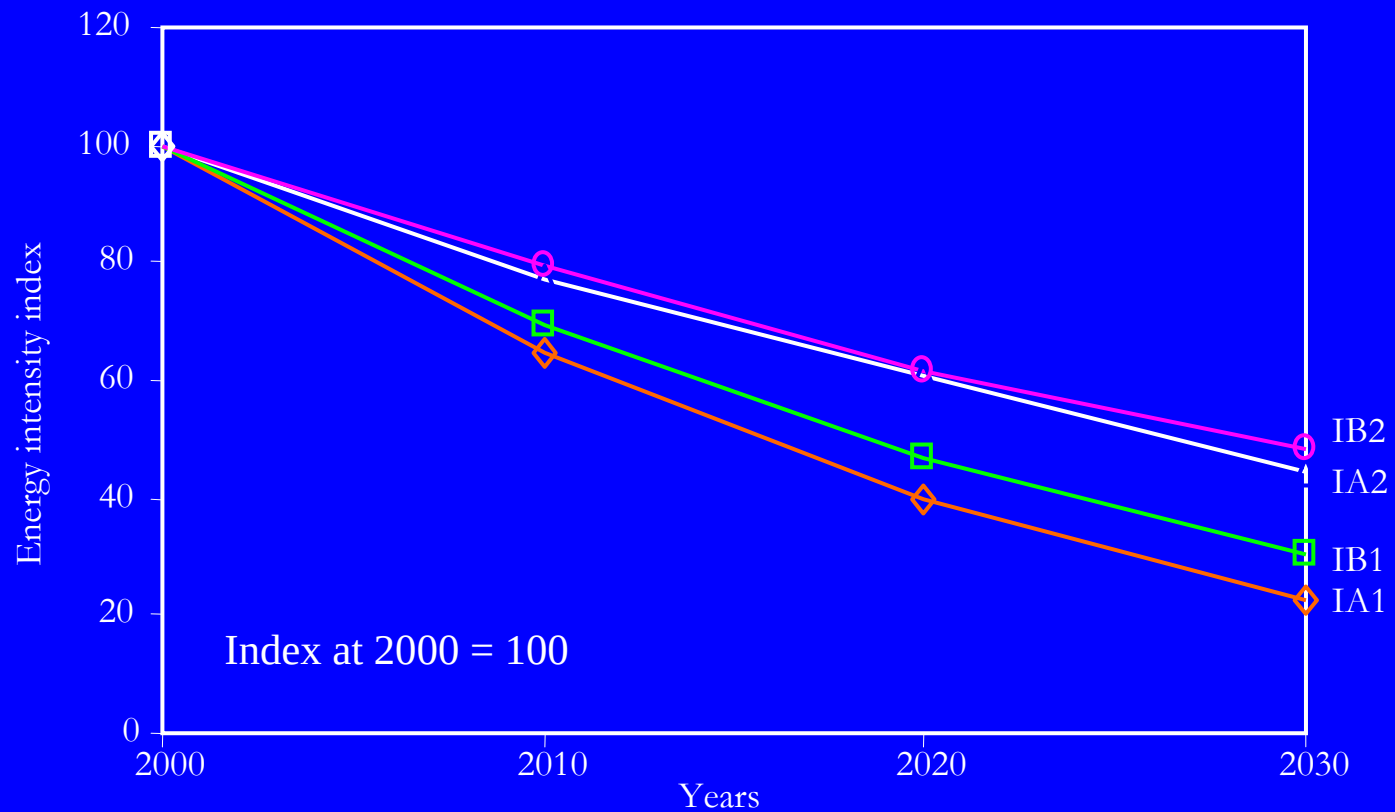
Energy Consumption (Mtoe)



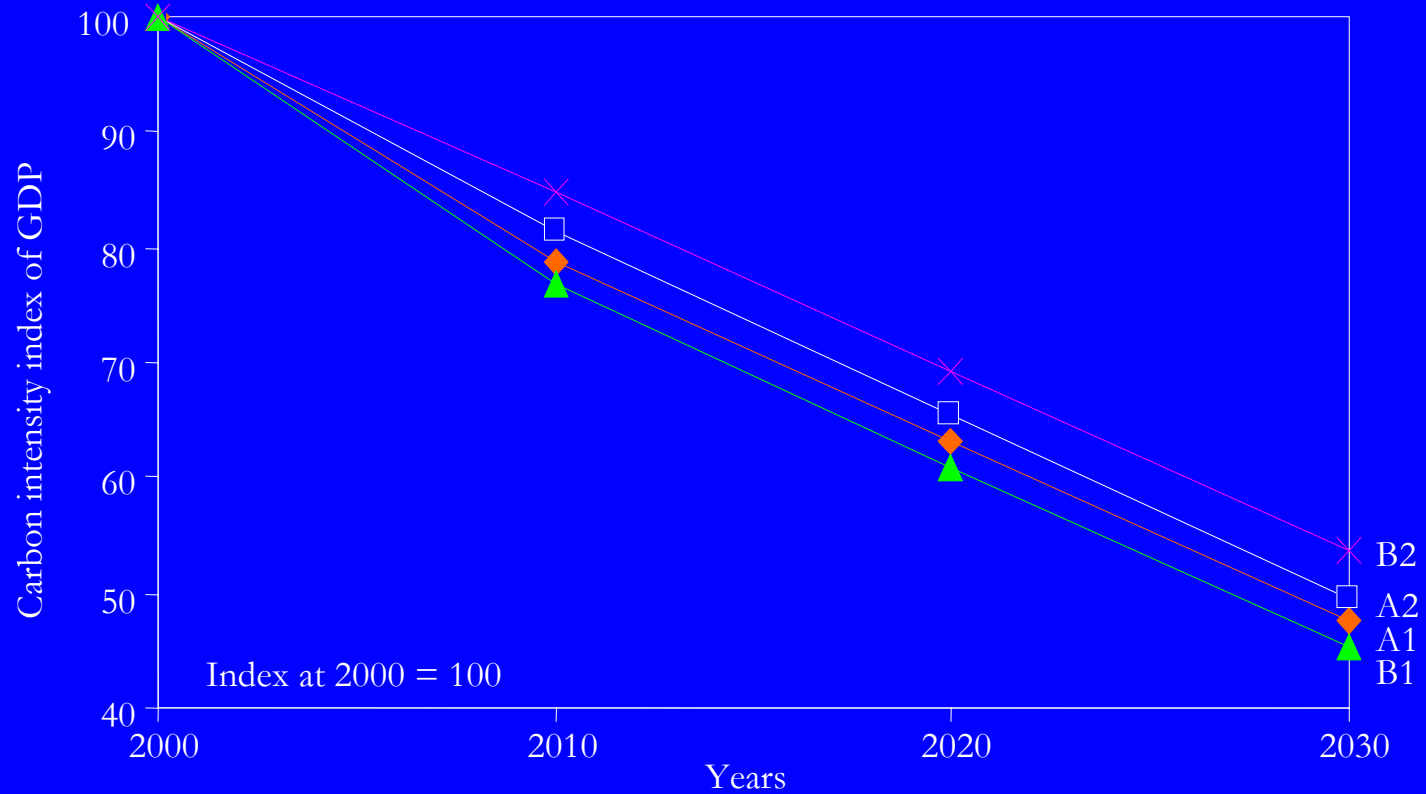
Carbon Emission (Mton)



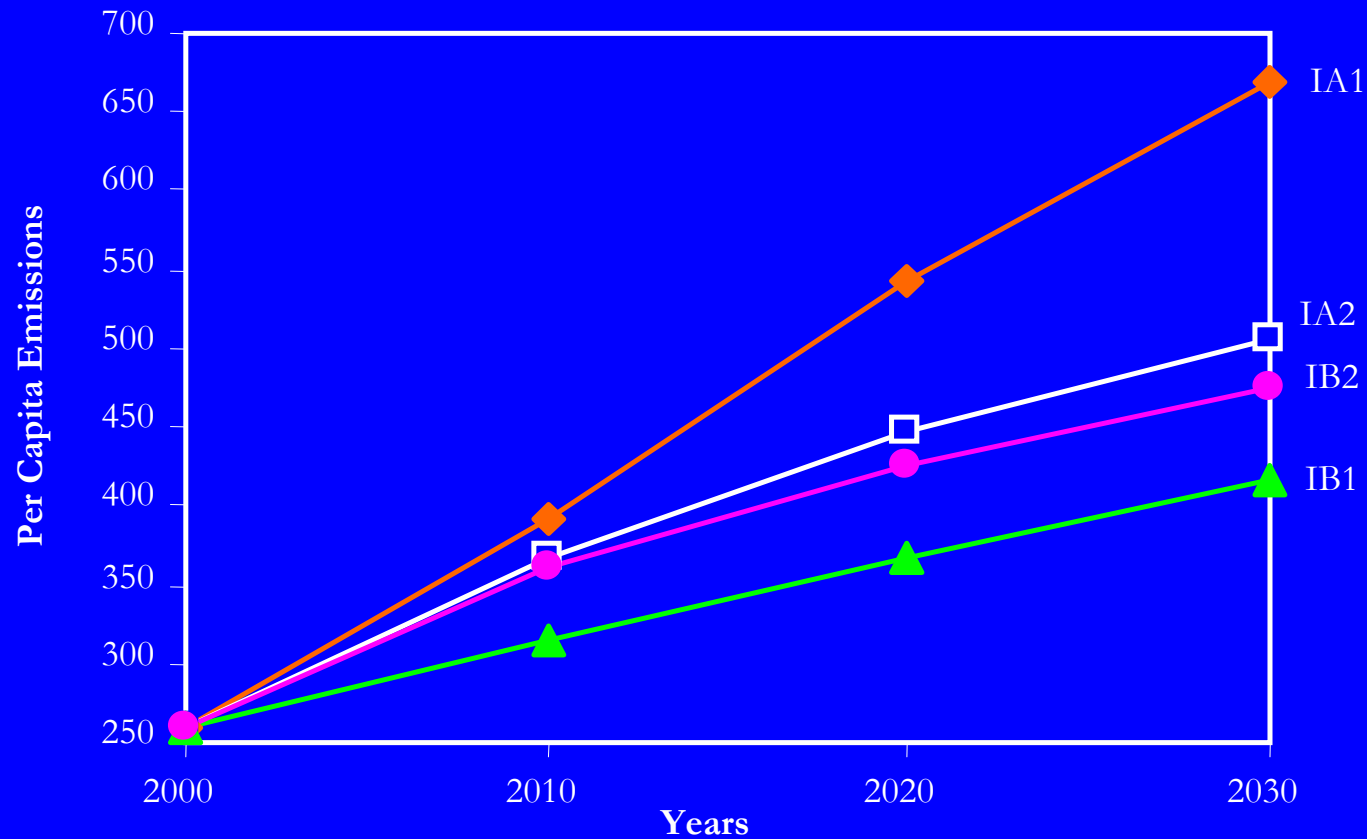
Energy Intensity



Carbon Intensity



Per Capita Carbon Emission (KgC/person)



Possible Consideration for Future GEO Scenario Framework specific to Asia and the Pacific Region

- Approach to building scenarios should be clear and transparent
- Access to time-series data
- SRES scenario approach should be explored
- Quantitative illustration of **Fortress World** and **Great Transitions** Scenarios
- Extension of time horizon to **2050** (GSG)

Thank You